

# Science

Nature Notebook  
Nature Walks & Scouting  
Physics Lessons  
Physics Labs





## About the Course

This course includes the following topic(s): Physics Lessons, Physics Labs, Nature Notebook: Grades 9-12, Nature Walks & Scouting: Grades 9-12

### About Nature Notebook: Grades 9-12

Outdoor work is established or continued as a lifelong habit. Optional resources are provided in science lessons and on the Alveary bookshelf.

### About Nature Walks & Scouting: Grades 9-12

Outdoor work is established or continued as a lifelong habit. Optional resources are provided in science lessons and on the Alveary bookshelf.

### About Physics Lessons

Physics students gain a solid, hands-on foundation in this living lecture series from Math without Borders. This course guides students through the process of challenging their established perceptions by building up new observations. The course provides extensive experience with technology and mathematical modeling for a stand-alone high school physics course or as a foundation for a future, calculus-based physics course.

### About Physics Labs

Labs are an essential part of science in which students engage with the Things they are reading about and practice the scientific method. Labs for this course are integrated into the lessons to facilitate adequate time for more involved activities and to better coordinate with the lessons.



## Placement & Combining Tips

### Nature Notebook: Grades 9-12

Learners may be combined and follow their own interests.

### Nature Walks & Scouting: Grades 9-12

Learners may follow their own interests or follow the plan of their local scouting troop or natural history club.

### Physics Lessons

Recommended for students in 11th or 12th Grade, depending on math placement. Students should have completed Algebra II/Trigonometry and preferably be taking at least Precalculus concurrently with this course.



## Scheduling

| GRADE | SCHEDULE INFO.                                              | BOOKS |
|-------|-------------------------------------------------------------|-------|
| 9-12  | Nature Notebook: Grades 9-12<br>1+ time/week 20 min+        |       |
| 9-12  | Nature Walks & Scouting: Grades 9-12<br>1 time/week 30 min+ |       |

| GRADE | SCHEDULE INFO.                         | BOOKS                                                               |
|-------|----------------------------------------|---------------------------------------------------------------------|
| 11-12 | Physics Lessons<br>5 times/week 45 min | Home Study Companion: Physics<br>Schaum's Outlines: College Physics |
| 11-12 | Physics Labs<br>1 time/week 60 min     |                                                                     |

#### Sample Weekly View

| Day 1                                                      | Day 2           | Day 3           | Day 4                                              | Day 5                           |
|------------------------------------------------------------|-----------------|-----------------|----------------------------------------------------|---------------------------------|
| <b>Science: Physics</b>                                    |                 |                 |                                                    |                                 |
| Physics Lessons<br>Nature Walks &<br>Scouting: Grades 9-12 | Physics Lessons | Physics Lessons | Physics Lessons<br>Nature Notebook:<br>Grades 9-12 | Physics Lessons<br>Physics Labs |



## Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

**LINKS:** Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

### Physics Lessons

- ☐ Carve out time to continue or establish the regular habit of spending time in nature, including the use of a nature journal, as appropriate.
- ☐ Obtain materials from the supply lists.
- ☐ Note that time stamps have been included for each video in your lesson plans, so that the lesson time can be better managed. Lectures that are a bit too long for a single lesson have been broken into parts.
- ☐ From the HSC Physics Unit 01 folder, review the PDFs Assessment, Installation, Course Overview, and License Agreement. We also recommend printing the Overview PDFs for each unit, as these provide an outline for the course. Keep these in your binder or folder.
- ☐ Shortcut/download Geogebra from the Quick Links.
- ☐ Download Tracker from the Quick Links.
- ☐ If your smartphone does not allow you to control shutter speed, download a camera app, such as ProShot.
- ☐ If you need a media player for this course, Mr. Chandler recommends the VLC media player. You may use whatever calculator you prefer.
- ☐ If you would like to use the same calculator as Mr. Chandler, download Free42 from the Quick Links. You may use whatever calculator you prefer.
- ☐ If you would like to use the same spreadsheet as Mr. Chandler, download Libre Office from the Quick Links. You may use whatever spreadsheet you prefer.
- ☐ Plan for discussion and engagement. Teachers who do not feel that they can discuss the

subject of this course with students should plan to either preread the material or complete the course alongside their students. Expertise is not necessary, but discussion is as important in math and science as in history and literature.

☐ Labs/activities are presented in a fully integrated state in which the instructor's recommendations to complete certain activities or practice problems after a specified lesson are incorporated. To use them in this way, it is helpful for the lab to be scheduled after the 5th lesson. If your lab time is earlier in the week and you will find it difficult to be flexible, plan to delay the start of the lesson schedule so that lab day occurs after the 5th lesson. For example, if your lab time is on Tuesday, you might begin lesson 1 on Wednesday of the first week of school rather than on Monday. Those wanting to use the labs/activities on a more flexible schedule can do so, but should be ready and able to adjust lesson connections.

☐ Select a science book from the Alveary bookshelf for personal reading time, as appropriate.

#### Physics Labs

☐ Preview science labs and purchase materials as indicated on the supply list. Print or shortcut any links, as desired.

### Term Prep & Teacher Tips

#### Physics Lessons

- ☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:
  - the instructor will suggest found items throughout the course
  - any cell phone with a camera



## Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

### Science: Physics

#### Physics Lessons



Home Study Companion: Physics



Schaum's Outlines: College Physics



## Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

∞ [View Supply List Details](#)

### Science: Physics



Cell Phone Tripod



Household Items - Science: Physics



## Quick Links

### Science: Physics

#### Physics Lessons

- ∞ [Lab Notebook Examples](#)
- ∞ [John Muir Laws • Nature Journaling Resources](#)
- ∞ [Alveary Bookshelf](#)
- ∞ [Seek app from iNaturalist](#)
- ∞ [SkyView Lite for iOS](#)
- ∞ [SkyView Lite for Android](#)
- ∞ [GeoGebra](#)
- ∞ [Tracker](#)
- ∞ [Free42](#)
- ∞ [Libre Office](#)

Click THIS text  
or scan the QR  
code for links.



# Science: Physics

## How To Approach



### Introduce

- Learners begin each lesson with their existing knowledge. If continuing or revisiting a topic, then recap.
- Take note of the chapter or video title, and consider how the day's topic connects back to prior topic(s).
- Look back at the book or lecture notes as needed. The outline of the book and even of the chapter is helpful in drawing out and connecting ideas. Learners practice making these connections as they proceed through the course.



### View

- View or do, as instructed in the lessons. This course is mostly a living lecture series, but there are many opportunities to practice.
- The supplemental text is mainly there to do practice problems. It is not necessary to do all of the practice problems - try a variety, learn from them, and move on when the learner is confident.
- The supplemental text can supply a concise review of the material when learners need it, but don't want to rewatch the videos again, or if they do not understand a word or concept. It is not a replacement for the videos, however.
- As they view the lectures, learners record ideas in a notebook or binder using outlines, diagrams, graphic organizers, or other methods (or combination of methods) that suit them. These recordings can be a helpful mechanism for remembering or a mini-narration to support understanding.



### Narrate

- Process the ideas of the lesson by explaining a concept, describing an object, retelling events, etc. Consider what the point of the lesson is before deciding what to do with it.
- Learners may use words, pictures, models, graphics, etc, to process and convey ideas in their own way.



### Discuss

- Consider together any thoughts, confusion, or concerns about the lesson.
- Note that learners may need to spend more than the allotted time engaging with or even repeating a lesson before moving on or as reinforcement at a later time. Adjust the pace as needed to feed the learner.

- Notice if there are any dates to keep for the Book of Centuries or quotes for the Commonplace Book.



## Connect

- Follow any extra links or pursue the Science and Society topics further, depending on learner needs and interest.

# Science: Physics

## How To Approach



### Introduce

- This course uses more open-ended 'projects' for lab, but every activity still has the same flow, which follows the scientific method.
- Learners begin with an introduction. How does the project relate to what they have learned so far, as well as any previous experience? What will they learn from the lab? This is how hypotheses are formed.
- Learners are not given a Procedure and must decide for themselves how to approach the lab. If they are unsure what to do, it can help to dialogue with someone as they work through the ideas.
- Once they have had a chance to think, learners compose a prelab narration to put these introductory ideas, hypotheses, and plans into their lab notebooks. These prelab narrations may seem short and even incomplete if learners are new to keeping a lab notebook, but that is okay.
- If learners have difficulty or are easily frustrated, then consider allowing a digital notebook for typing, a scribe, or the use of assistive technology, as appropriate.
- After they complete their prelab narration, learners should collect any materials, being sure to let teachers know if something is missing or to remind the teacher if something needs to be purchased at the store.



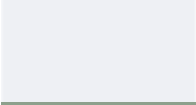
### Lab Procedure

- Learners complete the activity.
- They should use their lab notebooks to record what they do and capture any data, which can be printed from the computer and taped into the notebook.



### Analysis & Conclusions

- The last step in the lab is to analyze the data and observations and draw conclusions from them. Similar to the prelab narration, the concluding or postlab narration is a chance to think about and put into words. This time, learners are considering what they learned from the lab and what they could learn more about if they were to continue.
- Teachers should engage learners with this reflection by reviewing their lab notebooks with them, discussing the science used in the lab, and demonstrating curiosity about the lab themselves.
- Depending on the interest of the learner and the priorities of the teacher, the learner might be encouraged to spend more time on those ideas of what more they could learn or it might be time to move on. Either way, it is an important part of the scientific method to reflect on what we



could or would do next - our practice should help to clarify our thinking and teach that there is always more to be learned.

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## Term 1

### WEEK 1 ☐ 45m Physics Lessons - Lesson 1

☐ *45m Physics - Lesson 1*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → NOTE

These lesson plans are an organizational tool that can be used as much or as little as is helpful; the Home Study Companion is your course. Note that the times in parentheses indicate the length of the HSC videos mentioned.

#### → VIEW/READ, NARRATE, & DISCUSS

- Familiarize yourself with the new software you will be using.
- Print Unit overviews to keep in a folder for reference.
- Look over the PDF files in the Unit 1 folder.
- Read the Unit 1 overview.
- If needed, review helpful math concepts in the 1-00 Math Break folder (14.39, 1.30, 7.32) and Unit conversions and practice problems in video 1-01 (27.48). Some students may not need any of this review, while others might spend a few days here.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 1 ☐ 45m Physics Lessons - Lesson 2

☐ *45m Physics - Lesson 2*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Refer to the Unit 1 overview to note what video 1-02 is about before watching the lesson (25.27).
- Then, begin the Physics Problem Sets for Unit 1 with time remaining.

### WEEK 1 ☐ 45m Physics Lessons - Lesson 3

☐ *45m Physics - Lesson 3*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Refer to the Unit 1 overview to note what video 1-03 is about before watching the lesson (26.36).
- Then, continue Unit 1 Problem Set with the remaining time.

### WEEK 1 ☐ 45m Physics Lessons - Lesson 4

☐ *45m Physics - Lesson 4*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE/READ, NARRATE, & DISCUSS

- Complete Physics Problem Set for Unit 1.



## Term 1

- Check out 1-05 Science and Society, as desired.

### WEEK 1

#### 45m Physics Lessons - Lesson 5

##### 45m Physics - Lesson 5

Materials: Science Free Read, Math Without Borders Physics Home Study Companion, and computer access

PREP: Read Student/Teacher Tip

#### → READ, NARRATE, & DISCUSS

- Read science free read, as desired, then move on to the lab.

#### ★ STUDENT & TEACHER TIP

Depending on the learner and the schedule, it may work best to read during the lesson time and begin the lab later or to devote time to the lab first and read at a later time or another day. Notice what works best and adjust accordingly

### WEEK 1

#### 60m Physics Lessons - Lesson 6

##### 60m Physics Lab - Lesson 6

Materials: Math Without Borders Physics Home Study Companion and computer access

#### → LAB DAY

- If you need practice with measurements, do Project 1 in folder 1-04. Otherwise, watch the short video about Using Tracker in Project 2 (11.29), choose from among the given Projects, and get started.
- Be sure to write your introductory narration in your notebook and journal about your progress.
- When you finish, write your concluding narration.
- If you finish quickly, you can take a break for today or choose another Project to begin. If you don't finish today, you may do so another time. You do not need to do all of the Projects. You may continue them for fun in the afternoon or move on after doing one or two.

### WEEK 2

#### 45m Physics Lessons - Lesson 7

##### 45m Physics - Lesson 7

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Refer to the Unit 2 overview to note what video 2-01 is about before watching the lesson (32.05).

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 2

#### 45m Physics Lessons - Lesson 8

##### 45m Physics - Lesson 8

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Refer to the Unit 2 overview to note what video 2-02a is about before watching



## Term 1

the lesson (20.24).

- Then, watch 2-02b (16.52).

### WEEK 2 **45m Physics Lessons - Lesson 9**

 *45m Physics - Lesson 9*

 Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Refer to the Unit 2 overview to note what video 2-03a is about before watching the lesson (15.40).
- Then, begin the Physics Problem Sets for Unit 2 with time remaining.

### WEEK 2 **45m Physics Lessons - Lesson 10**

 *45m Physics - Lesson 10*

 Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch 2-04a (30.33).
- Then, continue the Problem Set for Unit 2.

### WEEK 2 **45m Physics Lessons - Lesson 11**

 *45m Physics - Lesson 11*

 Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch 2-04b (25.10).
- Then, continue the Problem Set for Unit 2.

### WEEK 2 **60m Physics Lessons - Lesson 12**

 *60m Physics Lab - Lesson 12*

 Materials: Math Without Borders Physics Home Study Companion and computer access

#### → LAB DAY

- Choose from among the Projects in folders 2-05 and start using your lab notebook to write your introductory narration, record your progress, and write your concluding narration.
- If you finish quickly, you can take a break for today or choose another Project to begin. If you don't finish today, you may do so another time (you will have more time next week for the Unit 2 projects). You do not need to do all of the Projects. You can continue them for fun in the afternoon or move on after doing one or two of them.



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### WEEK 3 45m Physics Lessons - Lesson 13

45m Physics - Lesson 13

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE

- Complete the Problem Set for Unit 2.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 3 45m Physics Lessons - Lesson 14

45m Physics - Lesson 14

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Refer to the Unit 3 overview to note what video 3-01a is about before watching the lesson (17.57).
- Use any remaining time to catch up on unfinished work or read from your science free read.

### WEEK 3 45m Physics Lessons - Lesson 15

45m Physics - Lesson 15

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Refer to the Unit 3 overview to note what video 3-02a is about before watching the lesson (27.00).

### WEEK 3 45m Physics Lessons - Lesson 16

45m Physics - Lesson 16

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Refer to the Unit 3 overview to note what video 3-03a is about before watching the lesson (15.51).
- Then, begin the Problem Set for Unit 3.

### WEEK 3 45m Physics Lessons - Lesson 17

45m Physics - Lesson 17

Materials: Math Without Borders Physics Home Study Companion, computer access, video link, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- ∞ Video Link: "Frames of Reference" (26.00)
- Then, move on to the lab.



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### WEEK 3 ☐ 60m Physics Lessons - Lesson 18

☐ *60m Physics Lab - Lesson 18*

☐ Materials: Math Without Borders Physics Home Study Companion and computer access

#### → LAB DAY

- Continue or try another Project from Unit 2 or move on to those in Unit 3, using your lab notebook to write your introductory narration, record your progress, and write your concluding narration.

### WEEK 4 ☐ 45m Physics Lessons - Lesson 19

☐ *45m Physics - Lesson 19*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Refer to the Unit 3 overview to note what video 3-04a is about before watching the lesson (21.57).
- Then, continue the Problem Set for Unit 3.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 4 ☐ 45m Physics Lessons - Lesson 20

☐ *45m Physics - Lesson 20*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 3-04b (26.59).
- Then, continue the Problem Set for Unit 3.

### WEEK 4 ☐ 45m Physics Lessons - Lesson 21

☐ *45m Physics - Lesson 21*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE/READ, NARRATE, & DISCUSS

- Complete Problem Set for Unit 4.
- Check out 3-06 Science and Society, as desired.

### WEEK 4 ☐ 45m Physics Lessons - Lesson 22

☐ *45m Physics - Lesson 22*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE/READ, NARRATE, & DISCUSS

- Refer to the Unit 4 overview before watching 4-01a (26.04).
- Then, begin a cumulative review with any time remaining.



## Term 1

### WEEK 4 ☐ 45m Physics Lessons - Lesson 23

☐ *45m Physics - Lesson 23*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Work some Problems from the Unit 4 set, as needed, then move on to the lab.

#### ! STUDENT & TEACHER REMINDER

Just like with previous readings, it may work best to work the problem set during this lesson time and begin the lab later or to devote time to the lab first and set aside time for problems later or on another day. Do what works best for the learner.

### WEEK 4 ☐ 60m Physics Lessons - Lesson 24

☐ *60m Physics Lab - Lesson 24*

☐ Materials: Math Without Borders Physics Home Study Companion and computer access

#### → LAB DAY

- Complete a Unit 3 Project, using your lab notebook to narrate and record.

### WEEK 5 ☐ 45m Physics Lessons - Lesson 25

☐ *45m Physics - Lesson 25*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE/READ, NARRATE, & DISCUSS

- Refer to the Unit 5 overview before watching video 5-00a (26.51).
- Then, catch up or review with any time remaining.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 5 ☐ 45m Physics Lessons - Lesson 26

☐ *45m Physics - Lesson 26*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE/READ, NARRATE, & DISCUSS

- Watch the lesson 5-01a (22.04).
- Then, catch up or review with any time remaining.

### WEEK 5 ☐ 45m Physics Lessons - Lesson 27

☐ *45m Physics - Lesson 27*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE/READ, NARRATE, & DISCUSS

- Watch the lesson 5-01b (17.39).
- Then, begin the Problem Set for Unit 5.



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### WEEK 5 ☐ 45m Physics Lessons - Lesson 28

☐ *45m Physics - Lesson 28*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Complete Problem Set for Unit 5.

### WEEK 5 ☐ 45m Physics Lessons - Lesson 29

☐ *45m Physics - Lesson 29*

☐ Materials: Science Free Read

Math Without Borders Physics Home Study Companion and computer access

#### → READ, NARRATE, & DISCUSS

- Read from your science free read, as desired, then move on to the lab.

### WEEK 5 ☐ 60m Physics Lessons - Lesson 30

☐ *60m Physics Lab - Lesson 30*

☐ Materials: Math Without Borders Physics Home Study Companion and computer access

#### → LAB DAY

- Complete a Unit 5 Project, using the lab notebook to narrate and record.

### WEEK 6 ☐ 45m Physics Lessons - Lesson 31

☐ *45m Physics - Lesson 31*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Refer to the Unit 6 overview before watching the lessons 6-01a (8.20) and 6-02a (23.00).

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 6 ☐ 45m Physics Lessons - Lesson 32

☐ *45m Physics - Lesson 32*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch the lesson 6-02b (23.23).
- Then, begin the Problem Set for Unit 6.



## Term 1

### WEEK 6 45m Physics Lessons - Lesson 33

45m Physics - Lesson 33

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch the lesson 6-02c (16.01).
- Then, continue the Problem Set for Unit 6.

### WEEK 6 45m Physics Lessons - Lesson 34

45m Physics - Lesson 34

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Refer to the Unit 6 overview before watching lesson 6-03a (16.18).
- Then, continue the Problem Set for Unit 6.

### WEEK 6 45m Physics Lessons - Lesson 35

45m Physics - Lesson 35

Materials: Science Free Read, Math Without Borders Physics Home Study Companion, and computer access

#### → VIEW/READ, NARRATE, & DISCUSS

- Check out 6-05 Science and Society or read from your science free read, as desired, then move on to the lab.

### WEEK 6 60m Physics Lessons - Lesson 36

60m Physics Lab - Lesson 36

Materials: Math Without Borders Physics Home Study Companion and computer access

#### → LAB DAY

- Complete a Unit 6 Project, using the lab notebook to narrate and record.

### WEEK 7 45m Physics Lessons - Lesson 37

45m Physics - Lesson 37

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Refer to the Unit 7 overview before watching lesson 7-01 (34.41).

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read



## Term 1

### WEEK 7 45m Physics Lessons - Lesson 38

45m Physics - Lesson 38

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Complete Problem Set for Unit 7.

### WEEK 7 45m Physics Lessons - Lesson 39

45m Physics - Lesson 39

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Refer to the Unit 8 overview before watching lessons 8-01a (10.16) and 8-02a (27.21).

### WEEK 7 45m Physics Lessons - Lesson 40

45m Physics - Lesson 40

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Refer to the Unit 8 overview before watching lesson 8-03a (21.46).
- Then, begin Problem Set for Unit 8 with any time remaining.
- Check out 8-06 Science and Society, as desired.

### WEEK 7 45m Physics Lessons - Lesson 41

45m Physics - Lesson 41

Materials: Science Free Read, Math Without Borders Physics Home Study Companion, and computer access

#### → READ, NARRATE, & DISCUSS

- Read from your science free read, as desired, then move on to the lab.

### WEEK 7 60m Physics Lessons - Lesson 42

60m Physics Lab - Lesson 42

Materials: Math Without Borders Physics Home Study Companion and computer access

#### → LAB DAY

- Complete a Unit 7 Project, using the lab notebook to narrate and record.



## Term 1

### WEEK 8 ☐ 45m Physics Lessons - Lesson 43

☐ *45m Physics - Lesson 43*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 8-03b (22.08).
- Then, begin/continue Problem Set for Unit 8.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 8 ☐ 45m Physics Lessons - Lesson 44

☐ *45m Physics - Lesson 44*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Refer to the Unit 8 overview before watching lesson 8-04a (29.59).
- Then, continue the Problem Set for Unit 8.

### WEEK 8 ☐ 45m Physics Lessons - Lesson 45

☐ *45m Physics - Lesson 45*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 8-04b (13.24).
- Then, continue the Problem Set for Unit 8.

### WEEK 8 ☐ 45m Physics Lessons - Lesson 46

☐ *45m Physics - Lesson 46*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Complete the Problem Set for Unit 8.
- Then, begin a cumulative review with any time remaining.

### WEEK 8 ☐ 45m Physics Lessons - Lesson 47

☐ *45m Physics - Lesson 47*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → READ, NARRATE, & DISCUSS

- Read from your science free read, as desired, then move on to the lab.



## Term 1

### WEEK 8 ☐ 60m Physics Lessons - Lesson 48

☐ *60m Physics Lab - Lesson 48*

☐ Materials: Math Without Borders Physics Home Study Companion and computer access

#### → LAB DAY

- Complete a Unit 8 project, using the lab notebook to narrate and record.

### WEEK 9 ☐ 45m Physics Lessons - Lesson 49

☐ *45m Physics - Lesson 49*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Refer to the Unit 9 overview before watching lessons 9-01a (20.06) and 9-02a (14.24).
- Then, continue the cumulative review with any time remaining.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 9 ☐ 45m Physics Lessons - Lesson 50

☐ *45m Physics - Lesson 50*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW, NARRATE, & DISCUSS

- Watch lesson 9-03a (37.56).

### WEEK 9 ☐ 45m Physics Lessons - Lesson 51

☐ *45m Physics: Lesson 51*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Begin Unit 9 Problem Set.

### WEEK 9 ☐ 45m Physics Lessons - Lesson 52

☐ *45m Physics - Lesson 52*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Complete Unit 9 Problem Set.



## Term 1

### WEEK 9 ☐ 45m Physics Lessons - Lesson 53

☐ 45m Physics - Lesson 53

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Refer to the Unit 10 overview before watching lesson 10-01 (24.59).

### WEEK 9 ☐ 60m Physics Lessons - Lesson 54

☐ 60m Physics Lab - Lesson 54

☐ Materials: Math Without Borders Physics Home Study Companion and computer access

#### → LAB DAY

- Complete a Unit 9 Project, using the lab notebook to narrate and record.

### WEEK 10 ☐ 45m Physics Lessons - Lesson 55

☐ 45m Physics - Lesson 55

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 10-02a (36.54).

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 10 ☐ 45m Physics Lessons - Lesson 56

☐ 45m Physics - Lesson 56

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 10-02b (35.10).

### WEEK 10 ☐ 45m Physics Lessons - Lesson 57

☐ 45m Physics - Lesson 57

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 10-02c (26.21).



## Term 1

### WEEK 10 ☐ 45m Physics Lessons - Lesson 58

☐ *45m Physics - Lesson 58*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 10-03a (32.54).

### WEEK 10 ☐ 45m Physics Lessons - Lesson 59

☐ *45m Physics - Lesson 59*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 10-03b (28.58), then move on to the lab with any time remaining.

### WEEK 10 ☐ 60m Physics Lessons - Lesson 60

☐ *60m Physics Lab - Lesson 60*

☐ Materials: Math Without Borders Physics Home Study Companion and computer access

#### → LAB DAY

- Complete a Unit 10 Project, using the lab notebook to narrate and record.

### WEEK 11 ☐ 45m Physics Lessons - Lesson 61

☐ *45m Physics Catch Up - Lesson 61*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, Schaum's College Physics, Link, and Science Free Read

#### → REPORT WRITING & REVIEW

- Spend this week writing a lab report on a physics lab of your choice from this term; review and edit the lab report before turning it in. A Sample Lab Report is provided in the link as an example. Use complete sentences, good grammar, 3rd person voice, significant figures, and scientific notation, as appropriate.
- Review what you have learned so far this term.
- Read from your Science Free Read.

∞ Link: Sample Lab Report

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read



## Term 1

### WEEK 11 ☐ 45m Physics Lessons - Lesson 62

☐ *45m Physics Catch Up - Lesson 62*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, Schaum's College Physics, Link, and Science Free Read

#### → REPORT WRITING & REVIEW

- Spend this week writing the lab report, reviewing what you have learned so far this term, and reading from your Science Free Read.

### WEEK 11 ☐ 45m Physics Lessons - Lesson 63

☐ *45m Physics Catch Up - Lesson 63*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, Schaum's College Physics, Link, and Science Free Read

#### → REPORT WRITING & REVIEW

- Spend this week writing the lab report, reviewing what you have learned so far this term, and reading from your Science Free Read.

### WEEK 11 ☐ 45m Physics Lessons - Lesson 64

☐ *45m Physics Catch Up - Lesson 64*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, Schaum's College Physics, Link, and Science Free Read

#### → REPORT WRITING & REVIEW

- Spend this week writing the lab report, reviewing what you have learned so far this term, and reading from your Science Free Read.

### WEEK 11 ☐ 45m Physics Lessons - Lesson 65

☐ *45m Physics Catch Up - Lesson 65*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, Schaum's College Physics, Link, and Science Free Read

#### → REPORT WRITING & REVIEW

- Spend this week writing the lab report, reviewing what you have learned so far this term, and reading from your Science Free Read.



## Term 1

### WEEK 11 ☐ 60m Physics Lessons - Lesson 66

☐ *60m Physics Catch Up - Lesson 66*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, Schaum's College Physics, Link, and Science Free Read

#### → REPORT WRITING & REVIEW

- Spend this week writing the lab report, reviewing what you have learned so far this term, and reading from your Science Free Read.

### WEEK 12 ☐ 45m Physics Lessons - Lesson 67

☐ *45m Physics Exams - Lesson 67*

Exam Week

#### → EXAMS

Answer question(s) related to course from Schaum's. Complete about 3 questions per day or as directed by your teacher. You may use reference material from your book regarding constants and equations, as needed.

### WEEK 12 ☐ 45m Physics Lessons - Lesson 68

☐ *45m Physics Exams - Lesson 68*

Exam Week

#### → EXAMS

Answer question(s) related to course from Schaum's.

### WEEK 12 ☐ 45m Physics Lessons - Lesson 69

☐ *45m Physics Exams - Lesson 69*

Exam Week

#### → EXAMS

Answer question(s) related to course from Schaum's.

### WEEK 12 ☐ 45m Physics Lessons - Lesson 70

☐ *45m Physics Exams - Lesson 70*

Exam Week

#### → EXAMS

Answer question(s) related to course from Schaum's.

### WEEK 12 ☐ 45m Physics Lessons - Lesson 71

☐ *45m Physics Exams - Lesson 71*

Exam Week

#### → EXAMS

Answer question(s) related to course from Schaum's.

# Science: Physics

[Click THIS text or scan the QR code for links.](#)



## Term 1

**WEEK 12** ☐ **60m Physics Lessons - Lesson 72**

☐ *60m Physics Exams - Lesson 72*

Exam Week

→ **EXAMS**

Answer question(s) related to course from Schaum's.



## Term 2

### WEEK 1 45m Physics Lessons - Lesson 73

45m Physics - Lesson 73

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Refer to the Unit 11 overview before watching lesson 11-01a to timestamp 20.35.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 1 45m Physics Lessons - Lesson 74

45m Physics - Lesson 74

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Complete lesson 11-01a from timestamp 20.35 (to 34.21).
- Then, begin Problem Set for Unit 11 (Ch.10 Schaum's).

### WEEK 1 45m Physics Lessons - Lesson 75

45m Physics - Lesson 75

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Begin lesson 11-02, watching to timestamp 22.43.
- Then, continue Problem Set for Unit 11 (Ch.10 Schaum's).

### WEEK 1 45m Physics Lessons - Lesson 76

45m Physics - Lesson 76

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Finish lesson 11-02 from timestamp 22.43 (to 41.41),
- Then, continue practice problems for Unit 11 (Ch.10 Schaum's).

### WEEK 1 45m Physics Lessons - Lesson 77

45m Physics - Lesson 77

Materials: Science Free Read, Math Without Borders Physics Home Study Companion, and computer access

#### → READ, NARRATE, & DISCUSS

- Read from your science free read, as desired, then move on to the lab.

#### ! STUDENT & TEACHER REMINDER

Continue to do what works best for the learner, reading during the lesson time and beginning the lab later or devoting time to the lab first and reading at a later time or another day.



## Term 2

### WEEK 1 ☐ 60m Physics Lessons - Lesson 78

☐ *60m Physics Lab - Lesson 78*

☐ Materials: Math Without Borders Physics Home Study Companion and computer access

#### → LAB DAY

- Complete a Unit 11 Project, using the lab notebook to narrate and record.

### WEEK 2 ☐ 45m Physics Lessons - Lesson 79

☐ *45m Physics - Lesson 79*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 11-03 (25.16).
- Then, continue Practice Problems for Unit 11 (Ch.10 Schaum's).

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 2 ☐ 45m Physics Lessons - Lesson 80

☐ *45m Physics - Lesson 80*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 11-04a (24.22).
- Then, continue Practice Problems for Unit 11 (Ch.10 Schaum's).

### WEEK 2 ☐ 45m Physics Lessons - Lesson 81

☐ *45m Physics - Lesson 81*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Continue Unit 11 Problem Set.

### WEEK 2 ☐ 45m Physics Lessons - Lesson 82

☐ *45m Physics - Lesson 82*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Complete Unit 11 Problem Set.



## Term 2

### WEEK 2 ☐ 45m Physics Lessons - Lesson 83

☐ 45m Physics - Lesson 83

☐ Materials: Science Free Read, Math Without Borders Physics Home Study Companion, and computer access

#### → READ, NARRATE, & DISCUSS

- Read from your science free read, as desired, then move on to the lab.

### WEEK 2 ☐ 60m Physics Lessons - Lesson 84

☐ 60m Physics Lab - Lesson 84

☐ Materials: Math Without Borders Physics Home Study Companion and computer access

#### → LAB DAY

- Complete a Unit 11 Project, using the lab notebook to narrate and record.

### WEEK 3 ☐ 45m Physics Lessons - Lesson 85

☐ 45m Physics - Lesson 85

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Refer to the Unit 12 overview before watching lesson 12-01 (24.49).

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 3 ☐ 45m Physics Lessons - Lesson 86

☐ 45m Physics - Lesson 86

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 12-02 (21.57).
- Then, begin the Problem Set for Unit 12 (Ch.11 Schaum's).

### WEEK 3 ☐ 45m Physics Lessons - Lesson 87

☐ 45m Physics - Lesson 87

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 12-03 (12.39).
- Then, continue the Problem Set for Unit 12 (Ch.11 Schaum's).



## Term 2

### WEEK 3 ☐ 45m Physics Lessons - Lesson 88

☐ 45m Physics - Lesson 88

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Complete Unit 12 Problem Set (Ch.11 Schaum's).

### WEEK 3 ☐ 45m Physics Lessons - Lesson 89

☐ 45m Physics - Lesson 89

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → READ, NARRATE, & DISCUSS

- Read from your science free read, as desired, then move on to the lab.

### WEEK 3 ☐ 60m Physics Lessons - Lesson 90

☐ 60m Physics Lab - Lesson 90

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → LAB DAY

- Complete a Unit 12, Part A Project, using the lab notebook to narrate and record.

### WEEK 4 ☐ 45m Physics Lessons - Lesson 91

☐ 45m Physics - Lesson 91

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Refer to the Unit 13 overview before watching lesson 13-01a (19.16).
- Then, begin a cumulative review with any time remaining.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 4 ☐ 45m Physics Lessons - Lesson 92

☐ 45m Physics - Lesson 92

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 13-02a (19.14).
- Then, continue a cumulative review with any time remaining.



## Term 2

### WEEK 4 ☐ 45m Physics Lessons - Lesson 93

☐ *45m Physics - Lesson 93*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 13-03a (20.23),
- Then, begin Problem Set for Unit 13 (Ch.22 Schaum's).

### WEEK 4 ☐ 45m Physics Lessons - Lesson 94

☐ *45m Physics - Lesson 94*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 13-04a (29.35).
- Then, continue Problem Set for Unit 13 (Ch.22 Schaum's).

### WEEK 4 ☐ 45m Physics Lessons - Lesson 95

☐ *45m Physics - Lesson 95*

☐ Materials: Science Free Read, Math Without Borders Physics Home Study Companion, and computer access

#### → READ, NARRATE, & DISCUSS

- Read from your science free read, as desired, then move on to the lab.

### WEEK 4 ☐ 60m Physics Lessons - Lesson 96

☐ *60m Physics Lab - Lesson 96*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → LAB DAY

- Complete a Project from Unit 12, Part A or Unit 13, using the lab notebook to narrate and record.

### WEEK 5 ☐ 45m Physics Lessons - Lesson 97

☐ *45m Physics - Lesson 97*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 13-05 (27.46).
- Then, continue Problem Set for Unit 13 (Ch.22 Schaum's).
- Check out 13-09 Science and Society, as desired.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read



## Term 2

### WEEK 5 45m Physics Lessons - Lesson 98

45m Physics - Lesson 98

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 13-06a (18.57).
- Then, continue Problem Set for Unit 13 (Ch.22 Schaum's).

### WEEK 5 45m Physics Lessons - Lesson 99

45m Physics - Lesson 99

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 13-07a (24.36).
- Then, continue Problem Set for Unit 13 (Ch.22 Schaum's).

### WEEK 5 45m Physics Lessons - Lesson 100

45m Physics - Lesson 100

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Continue Problem Set for Unit 13 (Ch.22 Schaum's).

### WEEK 5 45m Physics Lessons - Lesson 101

45m Physics - Lesson 101

Materials: Science Free Read, Math Without Borders Physics Home Study Companion, and computer access

#### → READ, NARRATE, & DISCUSS

- Read from your science free read, as desired, then move on to the lab.

### WEEK 5 60m Physics Lessons - Lesson 102

60m Physics Lab - Lesson 102

Materials: Math Without Borders Physics Home Study Companion and computer access

#### → LAB DAY

- Complete a Unit 13 Project, using the lab notebook to narrate and record.



## Term 2

### WEEK 6 45m Physics Lessons - Lesson 103

45m Physics - Lesson 103

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Complete Problem Set for Unit 13 (Ch.22 Schaum's).

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 6 45m Physics Lessons - Lesson 104

45m Physics - Lesson 104

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Refer to the Unit 14 overview before watching lesson 14-01a (25.54).
- Then, begin Problem Set for Unit 14 (Ch.36 Schaum's) with any time remaining.

### WEEK 6 45m Physics Lessons - Lesson 105

45m Physics - Lesson 105

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 14-02a (23.12).
- Then, continue Problem Set for Unit 14 (Ch.36 Schaum's).

### WEEK 6 45m Physics Lessons - Lesson 106

45m Physics - Lesson 106

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 14-02b (25.40).
- Then, continue Problem Set for Unit 14 (Ch.36 Schaum's).

### WEEK 6 45m Physics Lessons - Lesson 107

45m Physics - Lesson 107

Materials: Science Free Read, Math Without Borders Physics Home Study Companion, and computer access

#### → READ, NARRATE, & DISCUSS

- Read from your science free read, as desired, then move on to the lab.



## Term 2

### WEEK 6 ☐ 60m Physics Lessons - Lesson 108

☐ 60m Physics Lab - Lesson 108

☐ Materials: Math Without Borders Physics Home Study Companion and computer access

#### → LAB DAY

- Complete a Unit 14 project, using the lab notebook to narrate and record.

### WEEK 7 ☐ 45m Physics Lessons - Lesson 109

☐ 45m Physics - Lesson 109

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 14-03a (28.41).
- Then, continue Problem Set for Unit 14 (Ch.36 Schaum's) with any time remaining.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 7 ☐ 45m Physics Lessons - Lesson 110

☐ 45m Physics - Lesson 110

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 14-03b (17.00).
- Then, continue Problem Set for Unit 14 (Ch.36 Schaum's).

### WEEK 7 ☐ 45m Physics Lessons - Lesson 111

☐ 45m Physics - Lesson 111

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Complete Problem Set for Unit 14 (Ch.36 Schaum's).

### WEEK 7 ☐ 45m Physics Lessons - Lesson 112

☐ 45m Physics - Lesson 112

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Complete Problem Set for Unit 14 (Ch.36 Schaum's).



## Term 2

### WEEK 7 ☐ 45m Physics Lessons - Lesson 113

☐ 45m Physics - Lesson 113

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → READ, NARRATE, & DISCUSS

- Read from your science free read, as desired, then move on to the lab.

### WEEK 7 ☐ 60m Physics Lessons - Lesson 114

☐ 60m Physics Lab - Lesson 114

☐ Materials: Math Without Borders Physics Home Study Companion and computer access

#### → LAB DAY

- Complete a Unit 14 Project, using the lab notebook to narrate and record.

### WEEK 8 ☐ 45m Physics Lessons - Lesson 115

☐ 45m Physics - Lesson 115

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Refer to the Unit 15 overview before watching lesson 15-01a (10.36).
- Use any time remaining to catch up or review.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 8 ☐ 45m Physics Lessons - Lesson 116

☐ 45m Physics - Lesson 116

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 15-02a (32.51).

### WEEK 8 ☐ 45m Physics Lessons - Lesson 117

☐ 45m Physics - Lesson 117

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 15-02b (33.28).



## Term 2

|        |                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                              |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| WEEK 8 | <div> <div>45m Physics Lessons - Lesson 118</div> <div>45m Physics - Lesson 118</div> </div> <div> <div>Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics</div> </div> <div> <div>→ PRACTICE, NARRATE, &amp; DISCUSS</div> <div>Begin Problem Set for Unit 15 (Ch.37 Schaum's).</div> </div>         |                                                                                                                              |
| WEEK 8 | <div> <div>45m Physics Lessons - Lesson 119</div> <div>45m Physics - Lesson 119</div> </div> <div> <div>Materials: Science Free Read, Math Without Borders Physics Home Study Companion, computer access</div> </div> <div> <div>→ READ, NARRATE, &amp; DISCUSS</div> <div>Read from your science free read, as desired, then move on to the lab.</div> </div> |                                                                                                                              |
| WEEK 8 | <div> <div>60m Physics Lessons - Lesson 120</div> <div>60m Physics Lab - Lesson 120</div> </div> <div> <div>Materials: Math Without Borders Physics Home Study Companion and computer access</div> </div> <div> <div>→ LAB DAY</div> <div>Complete a Unit 15 project, using the lab notebook to narrate and record.</div> </div>                               |                                                                                                                              |
| WEEK 9 | <div> <div>45m Physics Lessons - Lesson 121</div> <div>45m Physics - Lesson 121</div> </div> <div> <div>Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics</div> </div> <div> <div>→ PRACTICE, NARRATE, &amp; DISCUSS</div> <div>Continue Problem Set for Unit 15 (Ch.37 Schaum's).</div> </div>      | <div> <div>• PLAN WEEKLY</div> <div> <div>nature walk - record observations</div> <div>science free read</div> </div> </div> |
| WEEK 9 | <div> <div>45m Physics Lessons - Lesson 122</div> <div>45m Physics - Lesson 122</div> </div> <div> <div>Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics</div> </div> <div> <div>→ PRACTICE, NARRATE, &amp; DISCUSS</div> <div>Complete Problem Set for Unit 15 (Ch.37 Schaum's).</div> </div>      |                                                                                                                              |



## Term 2

### WEEK 9 ☐ 45m Physics Lessons - Lesson 123

☐ *45m Physics - Lesson 123*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Refer to the Unit 16 overview before watching lessons 16-01a (15.05) and 16-02a (18.30).

### WEEK 9 ☐ 45m Physics Lessons - Lesson 124

☐ *45m Physics - Lesson 124*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 16-03a (28.47).
- Then, begin Practice Problems for Unit 16 (Ch.38-39 Schaum's) with any time remaining.

### WEEK 9 ☐ 45m Physics Lessons - Lesson 125

☐ *45m Physics - Lesson 125*

☐ Materials: Science Free Read, Math Without Borders Physics Home Study Companion, and computer access

#### → READ, NARRATE, & DISCUSS

- Read from your science free read, as desired, then move on to the lab.

### WEEK 9 ☐ 60m Physics Lessons - Lesson 126

☐ *60m Physics Lab - Lesson 126*

☐ Materials: Math Without Borders Physics Home Study Companion and computer access

#### → LAB DAY

- Complete a Unit 16 Project, using the lab notebook to narrate and record.

### WEEK 10 ☐ 45m Physics Lessons - Lesson 127

☐ *45m Physics - Lesson 127*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 16-04a (19.24).
- Then, continue Practice Problems for Unit 16 (Ch.38-39 Schaum's).

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read



## Term 2

### WEEK 10 ☐ 45m Physics Lessons - Lesson 128

☐ *45m Physics - Lesson 128*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Continue Practice Problems for Unit 16 (Ch.38-39 Schaum's).

### WEEK 10 ☐ 45m Physics Lessons - Lesson 129

☐ *45m Physics - Lesson 129*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Continue Practice Problems for Unit 16 (Ch.38-39 Schaum's).

### WEEK 10 ☐ 45m Physics Lessons - Lesson 130

☐ *45m Physics - Lesson 130*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Complete Practice Problems for Unit 16 (Ch.38-39 Schaum's).

### WEEK 10 ☐ 45m Physics Lessons - Lesson 131

☐ *45m Physics - Lesson 131*

☐ Materials: Science Free Read, Math Without Borders Physics Home Study Companion, and computer access

#### → READ, NARRATE, & DISCUSS

- Read from your science free read, as desired, then move on to the lab.

### WEEK 10 ☐ 60m Physics Lessons - Lesson 132

☐ *60m Physics Lab - Lesson 132*

☐ Materials: Math Without Borders Physics Home Study Companion and computer access

#### → LAB DAY

- Complete a Unit 16 project, using the lab notebook to narrate and record.



## Term 2

### WEEK 11 ☐ 45m Physics Lessons - Lesson 133

☐ 45m Physics Catch Up - Lesson 133

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, Schaum's College Physics, and Science Free Read

#### → REPORT WRITING & REVIEW

- Spend this week writing a lab report on a physics lab of your choice from this term; review and edit the lab report before turning it in. Use complete sentences, good grammar, 3rd person voice, significant figures, and scientific notation, as appropriate.
- Review what you have learned so far this term.
- Read from your Science Free Read.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 11 ☐ 45m Physics Lessons - Lesson 134

☐ 45m Physics Catch Up - Lesson 134

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, Schaum's College Physics, Link, and Science Free Read

#### → REPORT WRITING & REVIEW

- Spend this week writing the lab report, reviewing what you have learned so far this term, and reading from your Science Free Read.

### WEEK 11 ☐ 45m Physics Lessons - Lesson 135

☐ 45m Physics Catch Up - Lesson 135

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, Schaum's College Physics, Link, and Science Free Read

#### → REPORT WRITING & REVIEW

- Spend this week writing the lab report, reviewing what you have learned so far this term, and reading from your Science Free Read.

### WEEK 11 ☐ 45m Physics Lessons - Lesson 136

☐ 45m Physics Catch Up - Lesson 136

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, Schaum's College Physics, Link, and Science Free Read

#### → REPORT WRITING & REVIEW

- Spend this week writing a the lab report, reviewing what you have learned so far this term, and reading from your Science Free Read.



## Term 2

### WEEK 11 ☐ 45m Physics Lessons - Lesson 137

☐ *45m Physics Catch Up - Lesson 137*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, Schaum's College Physics, Link, and Science Free Read

#### → REPORT WRITING & REVIEW

- Spend this week writing the lab report, reviewing what you have learned so far this term, and reading from your Science Free Read.

### WEEK 11 ☐ 60m Physics Lessons - Lesson 138

☐ *60m Physics Catch Up - Lesson 138*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, Schaum's College Physics, Link, and Science Free Read

#### → REPORT WRITING & REVIEW

- Spend this week writing the lab report, reviewing what you have learned so far this term, and reading from your Science Free Read.

### WEEK 12 ☐ 45m Physics Lessons - Lesson 139

☐ *Physics Exams - Lesson 139*

Exam Week

#### → EXAMS

Answer question(s) related to course from Schaum's. Complete about 3 questions per day or as directed by your teacher. You may use reference material from your book regarding constants and equations, as needed.

### WEEK 12 ☐ 45m Physics Lessons - Lesson 140

☐ *45m Physics Exams - Lesson 140*

Exam Week

#### → EXAMS

Answer question(s) related to course from Schaum's.

### WEEK 12 ☐ 45m Physics Lessons - Lesson 141

☐ *45m Physics Exams - Lesson 141*

Exam Week

#### → EXAMS

Answer question(s) related to course from Schaum's.

### WEEK 12 ☐ 45m Physics Lessons - Lesson 142

☐ *45m Physics Exams - Lesson 142*

Exam Week

#### → EXAMS



| Term 2                                                                                                                                                                                                                                                            |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Answer question(s) related to course from Schaum's.                                                                                                                                                                                                               |  |
| <div><div>WEEK 12</div><div><div><div></div></div>45m Physics Lessons - Lesson 143</div><div><div><div></div></div>45m Physics Exams - Lesson 143</div><div>Exam Week</div><div>→ EXAMS</div><div>Answer question(s) related to course from Schaum's.</div></div> |  |
| <div><div>WEEK 12</div><div><div><div></div></div>60m Physics Lessons - Lesson 144</div><div><div><div></div></div>60m Physics Exams - Lesson 144</div><div>Exam Week</div><div>→ EXAMS</div><div>Answer question(s) related to course from Schaum's.</div></div> |  |



## Term 3

### WEEK 1 45m Physics Lessons - Lesson 145

45m Physics - Lesson 145

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Refer to the Unit 17 overview before watching lesson 17-01 (20.26).
- Then, look back over what you have learned this year with any time remaining.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 1 45m Physics Lessons - Lesson 146

45m Physics - Lesson 146

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lessons 17-02a (10.08) and 17-02b (15.56).

### WEEK 1 45m Physics Lessons - Lesson 147

45m Physics - Lesson 147

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 17-03a (12.41).
- Then, begin Practice Problems for Unit 17 (Ch.40 Schaum's) with any time remaining.

### WEEK 1 45m Physics Lessons - Lesson 148

45m Physics - Lesson 148

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 17-04a (26.31).
- Then, begin Practice Problems for Unit 17 (Ch.40 Schaum's) with any time remaining.

### WEEK 1 45m Physics Lessons - Lesson 149

45m Physics - Lesson 149

Materials: Science Free Read, Math Without Borders Physics Home Study Companion, and computer access

#### → READ, NARRATE, & DISCUSS

- Read from your science free read, as desired, then move on to the lab.



## Term 3

### WEEK 1 ☐ 60m Physics Lessons - Lesson 150

☐ *60m Physics Lab - Lesson 150*

☐ Materials: Math Without Borders Physics Home Study Companion and computer access

#### → LAB DAY

- Complete a Unit 17 Project, using the lab notebook to narrate and record.

### WEEK 2 ☐ 45m Physics Lessons - Lesson 151

☐ *45m Physics - Lesson 151*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 17-05a (27.44).
- Then, continue Practice Problems for Unit 17 (Ch.40 Schaum's) with any time remaining.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 2 ☐ 45m Physics Lessons - Lesson 152

☐ *45m Physics - Lesson 152*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Continue Practice Problems for Unit 17 (Ch.40 Schaum's).

### WEEK 2 ☐ 45m Physics Lessons - Lesson 153

☐ *45m Physics - Lesson 153*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Continue Practice Problems for Unit 17 (Ch.40 Schaum's).

### WEEK 2 ☐ 45m Physics Lessons - Lesson 154

☐ *45m Physics - Lesson 154*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Complete Practice Problems for Unit 17 (Ch.40 Schaum's).



## Term 3

### WEEK 2 ☐ 45m Physics Lessons - Lesson 155

☐ *45m Physics - Lesson 155*

☐ Materials: Science Free Read

Math Without Borders Physics Home Study Companion and computer access

#### → READ, NARRATE, & DISCUSS

- Read from your science free read, as desired, then move on to the lab.

### WEEK 2 ☐ 60m Physics Lessons - Lesson 156

☐ *60m Physics Lab - Lesson 156*

☐ Materials: Math Without Borders Physics Home Study Companion and computer access

#### → LAB DAY

- Complete a Unit 17 Project, using the lab notebook to narrate and record.

### WEEK 3 ☐ 45m Physics Lessons - Lesson 157

☐ *45m Physics - Lesson 157*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Refer to the Unit 18 overview before watching lesson 18-01a (23.19).
- Then, catch up or review with any time remaining.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 3 ☐ 45m Physics Lessons - Lesson 158

☐ *45m Physics - Lesson 158*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 18-01b (18.36).
- Then, catch up or review with any time remaining.

### WEEK 3 ☐ 45m Physics Lessons - Lesson 159

☐ *45m Physics - Lesson 159*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 18-02 (19.42).
- Then, begin Problem Set for Unit 18 (ch24 Schaum's).



## Term 3

### WEEK 3 ☐ 45m Physics Lessons - Lesson 160

☐ *45m Physics - Lesson 160*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 18-03 (30.47).
- Then, continue Problem Set for Unit 18 (ch24 Schaum's).

### WEEK 3 ☐ 45m Physics Lessons - Lesson 161

☐ *45m Physics - Lesson 161*

☐ Materials: Science Free Read, Math Without Borders Physics Home Study Companion, and computer access

#### → READ, NARRATE, & DISCUSS

- Read from your science free read, as desired, then move on to the lab.

### WEEK 3 ☐ 60m Physics Lessons - Lesson 162

☐ *60m Physics Lab - Lesson 162*

☐ Materials: Math Without Borders Physics Home Study Companion and computer access

#### → LAB DAY

- Complete a Unit 18 Project, using the lab notebook to narrate and record.

### WEEK 4 ☐ 45m Physics Lessons - Lesson 163

☐ *45m Physics - Lesson 163*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 18-04 (34.06).

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 4 ☐ 45m Physics Lessons - Lesson 164

☐ *45m Physics - Lesson 164*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Continue Problem Set for Unit 18 (Ch.24 Schaum's).



## Term 3

### WEEK 4 ☐ 45m Physics Lessons - Lesson 165

☐ *45m Physics - Lesson 165*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Continue Problem Set for Unit 18 (Ch.24 Schaum's).

### WEEK 4 ☐ 45m Physics Lessons - Lesson 166

☐ *45m Physics - Lesson 166*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Complete Problem Set for Unit 18 (Ch.24 Schaum's).

### WEEK 4 ☐ 45m Physics Lessons - Lesson 167

☐ *45m Physics the lab - Lesson 167*

☐ Materials: Math Without Borders Physics Home Study Companion and computer access

#### → READ & NARRATE

- Begin a Unit 18 Project, using the lab notebook to narrate and record.

### WEEK 4 ☐ 60m Physics Lessons - Lesson 168

☐ *60m Physics Lab - Lesson 168*

☐ Materials: Math Without Borders Physics Home Study Companion and computer access

#### → LAB DAY

- Complete a Unit 18 Project, using the lab notebook to narrate and record.

### WEEK 5 ☐ 45m Physics Lessons - Lesson 169

☐ *45m Physics - Lesson 169*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Refer to the Unit 19 overview before watching lesson 19-01a (31.20).

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read



## Term 3

### WEEK 5 45m Physics Lessons - Lesson 170

45m Physics - Lesson 170

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 19-01b (35.01).

### WEEK 5 45m Physics Lessons - Lesson 171

45m Physics - Lesson 171

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 19-02a (20.38).
- Then, begin Problem Set for Unit 19 (ch25 Schaum's).

### WEEK 5 45m Physics Lessons - Lesson 172

45m Physics - Lesson 172

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 19-02b (16.20).
- Then, continue Problem Set for Unit 19 (ch25 Schaum's).

### WEEK 5 45m Physics Lessons - Lesson 173

45m Physics the lab - Lesson 173

Materials: Math Without Borders Physics Home Study Companion and computer access

#### → READ & NARRATE

- Begin a Unit 19 Project, using the lab notebook to narrate and record.

### WEEK 5 60m Physics Lessons - Lesson 174

60m Physics Lab - Lesson 174

Materials: Math Without Borders Physics Home Study Companion and computer access

#### → LAB DAY

- Complete a Unit 19 Project, using the lab notebook to narrate and record.



## Term 3

### WEEK 6 45m Physics Lessons - Lesson 175

45m Physics - Lesson 175

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Continue Problem Set for Unit 19 (ch25 Schaum's).

#### • PLAN WEEKLY

- nature walk - record observations
- science free read

### WEEK 6 45m Physics Lessons - Lesson 176

45m Physics - Lesson 176

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Continue Problem Set for Unit 19 (ch25 Schaum's).

### WEEK 6 45m Physics Lessons - Lesson 177

45m Physics - Lesson 177

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Complete Problem Set for Unit 19 (ch25 Schaum's).

### WEEK 6 45m Physics Lessons - Lesson 178

45m Physics - Lesson 178

Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- In folder 20-00, refer to the Systems of Equations pdf.
- Then, watch lessons 4-08 (21.45) and 4-08 probs (18.07).

### WEEK 6 45m Physics Lessons - Lesson 179

45m Physics the lab - Lesson 179

Materials: Math Without Borders Physics Home Study Companion and computer access

#### → READ & NARRATE

- Begin a Unit 19 Project, using the lab notebook to narrate and record.



## Term 3

### WEEK 6 ☐ 60m Physics Lessons - Lesson 180

☐ *60m Physics Lab - Lesson 180*

☐ Materials: Math Without Borders Physics Home Study Companion and computer access

#### → LAB DAY

- Complete a Unit 19 Project, using the lab notebook to narrate and record.

### WEEK 7 ☐ 45m Physics Lessons - Lesson 181

☐ *45m Physics - Lesson 181*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 20-01a (21.47).
- Then, do a cumulative review with any time remaining.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 7 ☐ 45m Physics Lessons - Lesson 182

☐ *45m Physics - Lesson 182*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 20-01b (17.27).
- Then, do a cumulative review with any time remaining.

### WEEK 7 ☐ 45m Physics Lessons - Lesson 183

☐ *45m Physics - Lesson 183*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 20-02 (31.25).

### WEEK 7 ☐ 45m Physics Lessons - Lesson 184

☐ *45m Physics - Lesson 184*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 20-03 (23.19).
- Then, begin Problem Set for Unit 20 (Ch.26-27 Schaum's).



## Term 3

### WEEK 7 ☐ 45m Physics Lessons - Lesson 185

☐ *45m Physics the lab - Lesson 185*

☐ Materials: Math Without Borders Physics Home Study Companion and computer access

#### → READ & NARRATE

- Begin a Unit 20 project, using the lab notebook to narrate and record.

### WEEK 7 ☐ 60m Physics Lessons - Lesson 186

☐ *60m Physics Lab - Lesson 186*

☐ Materials: Math Without Borders Physics Home Study Companion and computer access

#### → LAB DAY

- Complete a Unit 20 project, using the lab notebook to narrate and record.
- Continue Problem Set for Unit 20 (Ch.26-27 Schaum's) with any time remaining.

### WEEK 8 ☐ 45m Physics Lessons - Lesson 187

☐ *45m Physics - Lesson 187*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 20-04a (21.59).
- Then, begin Problem Set for Unit 20 (Ch.26-27 Schaum's).

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 8 ☐ 45m Physics Lessons - Lesson 188

☐ *45m Physics - Lesson 188*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 20-04b (23.46).
- Then, begin Problem Set for Unit 20 (Ch.26-27 Schaum's).

### WEEK 8 ☐ 45m Physics Lessons - Lesson 189

☐ *45m Physics - Lesson 189*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Continue Problem Set for Unit 20 (Ch.26-27 Schaum's).



## Term 3

### WEEK 8 ☐ 45m Physics Lessons - Lesson 190

☐ 45m Physics - Lesson 190

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Complete Problem Set for Unit 20 (Ch.26-27 Schaum's).

### WEEK 8 ☐ 45m Physics Lessons - Lesson 191

☐ 45m Physics the lab - Lesson 191

☐ Materials: Math Without Borders Physics Home Study Companion and computer access

#### → READ & NARRATE

- Begin a Unit 20 Project, using the lab notebook to narrate and record.

### WEEK 8 ☐ 60m Physics Lessons - Lesson 192

☐ 60m Physics Lab - Lesson 192

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Science Free Read

#### → LAB DAY

- Complete a Unit 20 Project, using the lab notebook to narrate and record.
- Read Science Free Read with any time remaining.

### WEEK 9 ☐ 45m Physics Lessons - Lesson 193

☐ 45m Physics - Lesson 193

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Refer to the Unit 21 overview before watching lessons 21-01 (9.14) and 21-02 (30.27).

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 9 ☐ 45m Physics Lessons - Lesson 194

☐ 45m Physics - Lesson 194

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 21-03 (39.25).



## Term 3

### WEEK 9 ☐ 45m Physics Lessons - Lesson 195

☐ *45m Physics - Lesson 195*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → VIEW/READ, NARRATE, & DISCUSS

- Watch lesson 21-04 (34.32).

### WEEK 9 ☐ 45m Physics Lessons - Lesson 196

☐ *45m Physics - Lesson 196*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Begin Problem Set for Unit 21 (ch30-33 Schaum's).

### WEEK 9 ☐ 45m Physics Lessons - Lesson 197

☐ *45m Physics the lab - Lesson 197*

☐ Materials: Math Without Borders Physics Home Study Companion and computer access

#### → READ & NARRATE

- Begin a Unit 21 Project, using the lab notebook to narrate and record.

### WEEK 9 ☐ 60m Physics Lessons - Lesson 198

☐ *60m Physics Lab - Lesson 198*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Science Free Read

#### → LAB DAY

- Complete a Unit 21 Project, using the lab notebook to narrate and record.
- Read Science Free Read with any time remaining.

### WEEK 10 ☐ 45m Physics Lessons - Lesson 199

☐ *45m Physics - Lesson 199*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Continue Problem Set for Unit 21 (Ch.30-33 Schaum's).

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read



## Term 3

### WEEK 10 ☐ 45m Physics Lessons - Lesson 200

☐ *45m Physics - Lesson 200*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Continue Problem Set for Unit 21 (Ch.30-33 Schaum's).

### WEEK 10 ☐ 45m Physics Lessons - Lesson 201

☐ *45m Physics - Lesson 201*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Schaum's College Physics

#### → PRACTICE, NARRATE, & DISCUSS

- Complete Problem Set for Unit 21 (Ch.30-33 Schaum's).

### WEEK 10 ☐ 45m Physics Lessons - Lesson 202

☐ *45m Physics - Lesson 202*

☐ Materials: Science Free Read

#### → READ, NARRATE, & DISCUSS

Read your science free read and narrate or discuss what you read with your teacher.

### WEEK 10 ☐ 45m Physics Lessons - Lesson 203

☐ *45m Physics the lab - Lesson 203*

☐ Materials: Math Without Borders Physics Home Study Companion and computer access

#### → READ & NARRATE

- Begin a Unit 21 Project, using the lab notebook to narrate and record.

### WEEK 10 ☐ 60m Physics Lessons - Lesson 204

☐ *60m Physics Lab - Lesson 204*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, and Science Free Read

#### → LAB DAY

- Complete a Unit 21 Project, using the lab notebook to narrate and record.
- Read Science Free Read with any time remaining.



## Term 3

### WEEK 11 ☐ 45m Physics Lessons - Lesson 205

☐ *45m Physics Catch Up - Lesson 205*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, Schaum's College Physics, and Science Free Read

#### → REPORT WRITING & REVIEW

- Spend this week writing a lab report on a physics lab of your choice from this term; review and edit the lab report before turning it in. Use complete sentences, good grammar, 3rd person voice, significant figures, and scientific notation, as appropriate.
- Review what you have learned so far this term.
- Read from your Science Free Read.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 11 ☐ 45m Physics Lessons - Lesson 206

☐ *45m Physics Catch Up - Lesson 206*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, Schaum's College Physics, and Science Free Read

#### → REPORT WRITING & REVIEW

- Spend this week writing the lab report, reviewing what you have learned so far this term, and reading from your Science Free Read.

### WEEK 11 ☐ 45m Physics Lessons - Lesson 207

☐ *45m Physics Catch Up - Lesson 207*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, Schaum's College Physics, and Science Free Read

#### → REPORT WRITING & REVIEW

- Spend this week writing the lab report, reviewing what you have learned so far this term, and reading from your Science Free Read.

### WEEK 11 ☐ 45m Physics Lessons - Lesson 208

☐ *45m Physics Catch Up - Lesson 208*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, Schaum's College Physics, and Science Free Read

#### → REPORT WRITING & REVIEW

- Spend this week writing the lab report, reviewing what you have learned so far this term, and reading from your Science Free Read.



## Term 3

### WEEK 11 ☐ 45m Physics Lessons - Lesson 209

☐ *45m Physics Catch Up - Lesson 209*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, Schaum's College Physics, and Science Free Read

#### → REPORT WRITING & REVIEW

- Spend this week writing the lab report, reviewing what you have learned so far this term, and reading from your Science Free Read.

### WEEK 11 ☐ 60m Physics Lessons - Lesson 210

☐ *60m Physics Catch Up - Lesson 210*

☐ Materials: Math Without Borders Physics Home Study Companion, computer access, Schaum's College Physics, and Science Free Read

#### → REPORT WRITING & REVIEW

- Spend this week writing the lab report, reviewing what you have learned so far this term, and reading from your Science Free Read.

### WEEK 12 ☐ 45m Physics Lessons - Lesson 211

☐ *45m Physics Exams - Lesson 211*

Exam Week

#### → EXAMS

Answer question(s) related to course from Schaum's. Complete about 3 questions per day or as directed by your teacher. You may use reference material from your book regarding constants and equations, as needed.

### WEEK 12 ☐ 45m Physics Lessons - Lesson 212

☐ *45m Physics Exams - Lesson 212*

Exam Week

#### → EXAMS

Answer question(s) related to course from Schaum's.

### WEEK 12 ☐ 45m Physics Lessons - Lesson 213

☐ *45m Physics Exams - Lesson 213*

Exam Week

#### → EXAMS

Answer question(s) related to course from Schaum's.

### WEEK 12 ☐ 45m Physics Lessons - Lesson 214

☐ *45m Physics Exams - Lesson 214*

Exam Week

#### → EXAMS

Answer question(s) related to course from Schaum's.



| Term 3                                                                                                                                                                                                                                                                       |  |
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| <div><div>WEEK 12</div><div><div><div></div></div>45m Physics Lessons - Lesson 215</div><div><div><div></div></div>45m Physics Exams - Lesson 215</div><div>Exam Week</div><div><div>→ EXAMS</div><div>Answer question(s) related to course from Schaum's.</div></div></div> |  |
| <div><div>WEEK 12</div><div><div><div></div></div>60m Physics Lessons - Lesson 216</div><div><div><div></div></div>60m Physics Exams - Lesson 216</div><div>Exam Week</div><div><div>→ EXAMS</div><div>Answer question(s) related to course from Schaum's.</div></div></div> |  |

# Science: Physics

## Examination

### Term 1

#### Physics Lessons

- Schaum's Outlines: College Physics: Day 1: Choose 3 of the following from Schaum's:
  - #1.18, 1.26, 2.30, or 2.37
- Schaum's Outlines: College Physics: Day 2: Choose 3 of the following from Schaum's (substitute previous day's questions as appropriate based on material covered):
  - #3.50, 4.15, 5.16, or 6.33
- Schaum's Outlines: College Physics: Day 3: Choose 3 of the following from Schaum's (substitute previous days' questions as appropriate based on material covered):
  - #6.52, 8.34, 9.25, or 9.35
- Day 4: Explain one of your laboratory exercises from this term and what you learned from it. Describe any questions this made you think of.
- Day 5: Explain an example of physics in the world around you that you noticed this term. This might be from your project work, current events, nature walks, or your free read. Describe any questions this made you think of.

### Term 2

#### Physics Lessons

- Schaum's Outlines: College Physics: Day 1: Choose 3 of the following from Schaum's (substitute previous term's questions as appropriate based on material covered):
  - #10.40, 10.39, 11.21, or 11.29
- Schaum's Outlines: College Physics: Day 2: Choose 3 of the following from Schaum's (substitute previous questions as appropriate based on material covered):
  - #22.21, 22.27, 36.20, or 36.28
- Schaum's Outlines: College Physics: Day 3: Choose 3 of the following from Schaum's (substitute previous questions as appropriate based on material covered):
  - #37.15, 37.22, 38.26, or 39.25
- Day 4: Explain one of your laboratory exercises from this term and what you learned from it. Describe any questions this made you think of.
- Day 5: Explain an example of physics in the world around you that you noticed this term. This might be from your project work, current events, nature walks, or your free read. Describe any questions this made you think of.

## Term 3

### Physics Lessons

· Schaum's Outlines: College Physics: Day 1: Choose 3 of the following from Schaum's (substitute previous questions as appropriate based on material covered):

- #24.20, 24.24, 25.40, or 25.49

· Schaum's Outlines: College Physics: Day 2: Choose 3 of the following from Schaum's (substitute previous questions as appropriate based on material covered):

- #26.30, 26.36, 27.11, or 27.17

· Schaum's Outlines: College Physics: Day 3: Choose 3 of the following from Schaum's (substitute previous questions as appropriate based on material covered):

- #30.15, 31.14, 40.27, or 40.31

· Day 4: Explain one of your laboratory exercises from this term and what you learned from it. Describe any questions this made you think of.

· Day 5: Explain an example of physics in the world around you that you noticed this term. This might be from your project work, current events, nature walks, or your free read. Describe any questions this made you think of.